

B.Sc. PHYSICS · DEPARTMENT OF NATURAL SCIENCES · CUEA

Measuring the External Field

A cortical-proxy content observable and the
mean-field bound on media-driven opinion change

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Statistical physics has always modelled media as a **field** pushing on a population of coupled opinions.

That field is assumed. It is not measured.

This project measures one from content itself, applies it to 400 articles, and derives how strong the coupling would have to be for a spread that size to move a population at all.

01

The problem

What is missing from the way media is modelled.

Two articles can report one event and leave readers in different states

WIRE REPORT

Federal Reserve holds interest rates steady, citing a stable inflation outlook.

informs

THE SAME EVENT

FED DESTROYS AMERICA.
Your savings are GONE. The collapse they hid from you!

provokes

Readers notice the difference and cannot quantify it. The platforms distributing both have no objective measure of which they are amplifying.

Existing tools measure the words; none measures the predicted response



A released neural encoder predicts cortical activity from the content, and the measurement is taken on that prediction.

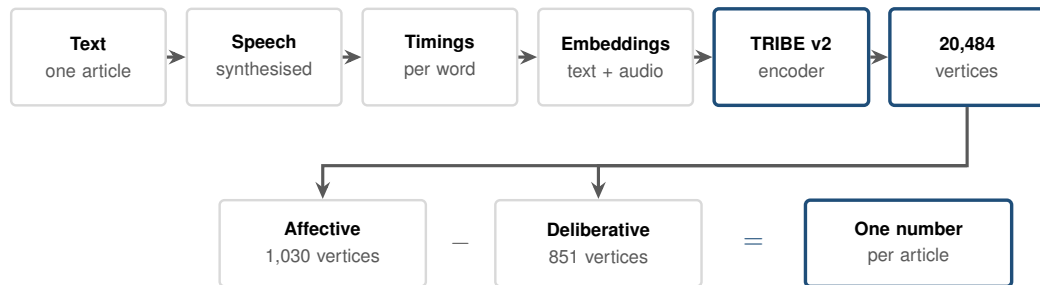
The instrument is a thermometer for one property of media. The thesis is an account of building it, running it, and stating what its readings support.

02

The instrument

From a passage of text to a single number.

A five-stage cascade turns one article into one number



Encoder: TRIBES v2, d'Ascoli et al., Meta FAIR (2025), arXiv:2507.22229. Regions from HCP MMP1.0, Glasser et al. (2016). Instrument at monarch-4iy.pages.dev.

The index contrasts two cortical networks, 1,030 vertices against 851



Affective salience in orange, deliberative control in blue. Resolved once into vertex index sets and cached, so all 400 items are averaged over identical regions.

Regions defined on HCP MMP1.0 (Glasser et al., 2016); surfaces are fsaverage5.

The instrument

The observable is a signed difference, defined for every item

$$NAA_{\text{signed}} = A_{\text{aff}} - A_{\text{del}}$$



A **cortical proxy**: it rates content, never a person, and is never called the amygdala.

Inspecting the checkpoint changed what the project could claim

01

Cortical only

No subcortical output.

No amygdala, so the proposal's equation cannot be computed.

02

Averaged over subjects

The loader forces it.

It predicts a typical viewer, which Paper 3 tests.

03

Standardised output

Means sit near zero, so a *ratio* breaks.

Undefined for 69 of 400 items; the difference replaces it.

All three were found by testing the checkpoint, before any result.

03

The measurement

400 articles, scanned twice.

The corpus is length-matched and was powered before the scan

100 fear-activating ISOT-fake	100 high outrage SemEval-2019 T4	100 reward hook Webis-Clickbait-17	100 neutral informational PubMed + ISOT-true
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Length-matched. Mean word counts 167.4, 163.5, 163.7, 162.2; standard deviations near 11.

Powered first. Smallest detectable $\eta^2 = 0.0268$, smallest detectable AUC = 0.5916, both fixed before a single item was scanned.

Corpora: Ahmed et al. (ISOT, 2017); Kiesel et al. (SemEval-2019 Task 4); Potthast et al. (Webis-Clickbait-17); PubMed open abstracts.

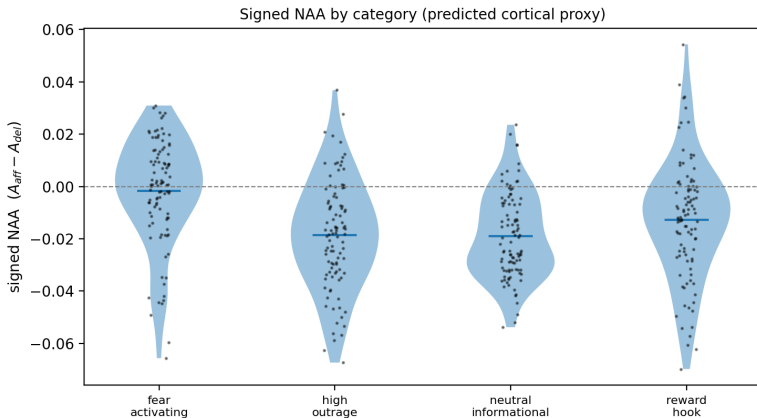
THE FOUR CATEGORIES SEPARATE AT

$$\eta^2 = 0.1068$$

$$F = 15.779, \quad p = 1.03 \times 10^{-9}, \quad n = 400$$

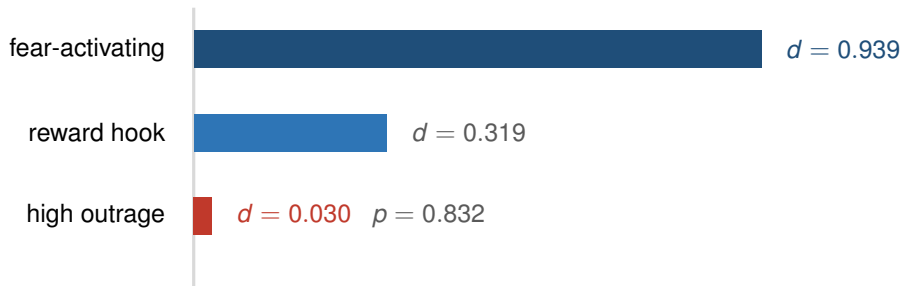
against a design powered to detect 0.0268

Categories shift against each other, but the distributions overlap



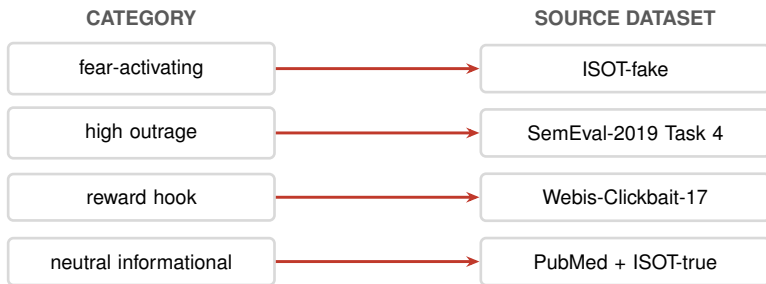
Every category mean is negative: the deliberative network leads throughout, and the categories differ in how far below zero they sit.

Fear-activating content carries the effect; outrage moves both networks equally



Cohen's d against the neutral baseline. **Outrage is the category the proposal expected to separate most**, and on the index it does not. Taken apart, it raises both networks: affective $d = +0.507$, deliberative $d = +0.491$. Fear raises one only.

Category cannot be told apart from source dataset



One source per category, so the separation is between *corpora* and cannot be attributed to *framing*. Length is controlled; provenance is not.

A bag of words beats the index on the label, so the index is not a classifier

0.6274

AUC, the index, nothing fitted

0.9758

AUC, TF-IDF + logistic, 5-fold

Same 400 rows, same label. Sentiment (VADER) sits at chance, 0.5392. The same words guess the *source* 66% of the time against 25% chance, so much of the lexical win is provenance.

The index is a field observable for the physics, not a detector.

Hutto & Gilbert (2014), VADER.

Group-level claims hold at this precision; per-item claims do not

0.8725

ICC between two GPU sessions

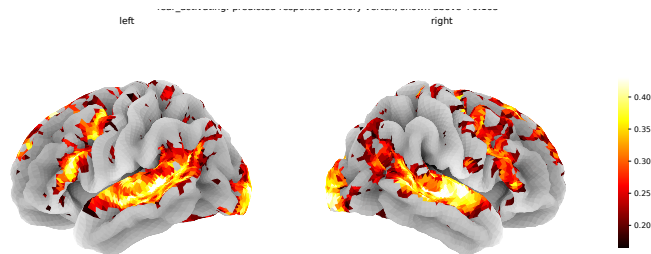
The separation replicates: $\eta^2 = 0.0888$
against 0.1068.

12.8%

of items reverse direction

51 of 400. Noise alone predicts 55, so no
verdict on one article is ever made.

The encoder returns a full cortical map, not a summary statistic



One fear-activating item at all 20,484 vertices, painted above the map's 70th percentile.
Nothing is smoothed, interpolated or simulated.

04

The physics

What a measured field is worth.

Media enters the mean-field model as a field on the measured observable

Each person holds an opinion ± 1 , is pulled toward neighbours by J , and pushed by media with field $h = \alpha X$, where X is the measured observable:

$$m = \tanh(\beta J m + h)$$

$$F(m) = am^2 + bm^4 - hm, \quad a = \frac{1 - \beta J}{2}, \quad b = \frac{1}{12}$$

Coefficients derived, not assumed; this corrects the proposal's a and b . The solver recovers the mean-field exponents.

The corpus does not identify the coupling, so no value is quoted

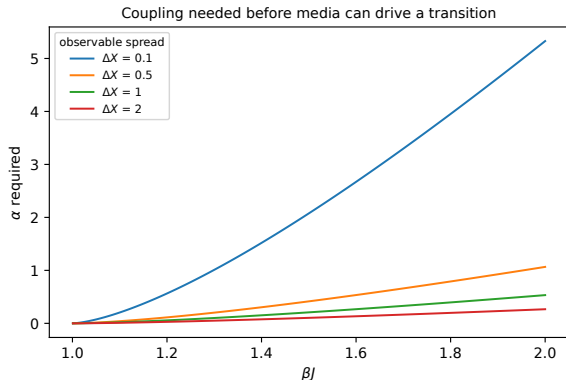
Tried. Fit the outcome to the measured index and read off α .

Result. Out of sample the fit does *worse than the average*, and the estimate depends on βJ , which nothing here measures.

So no value of α is quoted anywhere.

What replaces it needs no fit at all: the bound on the next slide.

Any content observable must clear a bound to drive a transition



$$\alpha \geq \frac{h_c(\beta J)}{\Delta X}$$

What α *would have to be* for media to tip opinion, for any observable, before any data. Within the mean-field model.

Measured $\Delta X = 0.1241$ gives
 $\alpha \geq 4.29$ at $\beta J = 2$.

05

Standing

What holds, and what is still open.

Against real brains the encoder is weakly positive, not anti-correlated

A company audit says the checkpoint this project runs is **anti-correlated** with real cortex. Tested here on public fMRI:

0.1517
noise ceiling
full episode, 4 subjects

+0.028
encoder vs brains
ceiling 0.096 on the same
80 scans, $p = 0.048$

3 bugs
silent, found, fixed
wrong episode, flipped
array, two clocks

Positive, weak, and not yet decisive on two minutes of film.

Algonauts 2025; Schaefer et al. (2018); audit: The Sapient Company (2026).

Four claims the work supports, and four it does not

SUPPORTED

- A content observable that can be measured, with its reliability stated.
- That it separates these four corpora at $\eta^2 = 0.1068$, power attached.
- A bound any candidate observable must satisfy, usable before data exists.
- A noise ceiling on public fMRI, and a first check of the encoder against it.

NOT SUPPORTED

- That the index detects manipulation. Source is confounded; TF-IDF scores 0.9758.
- Anything about the amygdala. The checkpoint does not predict it.
- Any value of the coupling α . The corpus does not identify it.
- Any verdict on one article. 12.8% reverse between sessions.

The work yields a dissertation and three papers

81 pp

Dissertation

Complete, review answered.

12 pp

Paper 1

The bound and the screening criterion.
Physica A. Independent of the scan.

16 pp

Paper 2

The instrument, the corpus, the field bound. *Physica A*.

8 pp

Paper 3

The noise ceiling and a check of the encoder against real brains.
Imaging Neuroscience.

Every number in all four comes from a re-runnable script. Nothing is transcribed by hand.

Conclusions

1. **A field observable can be measured from content.** The instrument runs, the corpus is complete, and reliability is measured rather than assumed: $ICC = 0.8725$.
2. **It separates four corpora at $\eta^2 = 0.1068$, and the separation is confounded with source.** Both halves are reported, the confound first.
3. **The coupling is unidentified, so the thesis reports a bound instead.** With $\Delta X = 0.1241$, $\alpha \geq 4.294$ at $\beta J = 2$ — a constraint on any future proposal of this mechanism.
4. **The encoder is weakly positive against real brains, not anti-correlated.** $r = +0.028$ against a ceiling of 0.096 on the same scans; not yet decisive.

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